Prompt-Driven Interactive Segmentation
as a Dataset-Annotation Tool

Final Report

Student A: 2301498
Abdelhafeez Aref Abdelhafeez Almaz

Student B: 2501925
Abdelrahman Eid Ashour Mohamed

Milestone 3 — November 1, 2026
CSE 681 — Digital Image Processing

Abstract

Manual instance-mask labeling is labor-intensive, motivating interactive segmentation tools that use a small number of human prompts. We implemented a local full-stack annotation application with a React/TypeScript frontend and FastAPI backend, using a frozen SAM 2.1 Hiera-Tiny checkpoint. The reproducible data workflow prepares 300 images from COCO 2017 val2017, with 979 non-crowd instances in 80 categories. The application supports positive and negative points, boxes, saving accepted masks, and computing IoU and Dice against the selected COCO instance. Image annotation supports multiple independently prompted objects, and an optional video workflow tracks multiple objects with independent prompts. A real API smoke test on Apple MPS returned a 428×640 mask with model confidence 0.954; the first request took 1.56 seconds including predictor initialization. However, no manual polygon baseline was measured and no prompted mask was scored in the available evaluation manifest. An additional video mode was smoke-tested with two independently propagated objects, including prompts on different frames, but not evaluated as an annotation experiment. Thus, this work establishes a runnable annotation and measurement pipeline, but does not establish accuracy, time savings, or superiority over manual annotation under the tested local setup.

1 Introduction

Creating instance masks by hand requires annotators to delineate object boundaries, which can be costly when preparing image datasets. Interactive segmentation offers a different workflow: an annotator provides points or a box, inspects a model-generated mask, and decides whether to refine or accept it. This project set out to build such a workflow around SAM 2 and evaluate it against existing COCO instance masks and a manual polygon baseline.

The deliverable is a local web application backed by Python/FastAPI and a React/TypeScript frontend. It provides a reproducible 300-image COCO 2017 validation subset, target-object selection, point and box prompts, local SAM 2 inference, accepted-mask persistence, and IoU/Dice computation. The software path was validated with automated tests and one real API inference on Apple MPS. That inference took 1.56 seconds on the first request, including model initialization, and returned model confidence 0.954. These are engineering and smoke-test observations, not comparative evaluation results. The planned manual polygon experiment and ground-truth-scored prompted sample were not completed, so the evidence does not support a claim that the tool improves mask quality or annotation speed.

2 Related Work

Interactive segmentation. GrabCut is a classic foreground-extraction technique in which user-supplied initialization guides iterative graph-cut refinement (Rother et al., 2004). This project follows the broader human-guided segmentation idea, but uses a promptable pretrained model rather than a graph-cut optimization.

Promptable foundation models. The Segment Anything Model introduced a large-scale promptable segmentation approach with point and box inputs (Kirillov et al., 2023). SAM 2 extends this family to image and video segmentation (Ravi et al., 2024); this project uses its image predictor with the Hiera-Tiny checkpoint and does not train or fine-tune the model. FastSAM is another recent approach to segment-anything behavior, using a different model pipeline (Zhao et al., 2023). These works provide context rather than directly comparable performance numbers: our smoke-test protocol and sample are not a benchmark reproduction.

Dataset and evaluation context. COCO supplies images and instance annotations for object recognition and segmentation research (Lin et al., 2014). We use its existing instance masks as references for post-annotation scoring. Each source image remains subject to its individual license and the dataset terms (COCO Consortium).

3 Method

The frozen SAM 2.1 Hiera-Tiny image predictor is wrapped by a FastAPI service. The React/TypeScript frontend lists local COCO images and their non-crowd object categories, serves a selected image, and provides positive point, negative point, and rectangle prompts for multiple objects. A segmentation request is made for each prompted object. The backend validates prompts against image dimensions, runs the image predictor, and returns the highest-scoring mask among the predictor's multimask candidates, together with its model confidence. The annotator can inspect color-coded masks and accept the selected object for COCO scoring.

The application also has an optional video mode. The browser accepts MP4 files up to 100 MiB, including longer videos, and lets the annotator choose a segment of up to 15 seconds. It samples up to 30 frames at 2 frames per second and resizes each sampled frame to a maximum long edge of 1280 pixels and 921,600 total pixels. Annotators can add point or box prompts on multiple objects and sampled frames and run SAM 2 video propagation independently in both directions for each object. This keeps masks and prompts separate and avoids coupling the objects' keyframe timing. The interface supports adding/removing object instances and editing each instance's point and rectangle prompts. Returned per-object binary masks can be downloaded as PNG files. This mode was verified with a short local smoke-test clip but was not part of a quality or timing experiment; its outputs are not scored against COCO.

For an accepted COCO annotation, the backend validates that the target annotation belongs to the selected image, checks the prediction dimensions, loads the corresponding exported ground-truth mask, and computes binary IoU and Dice. It saves the prediction mask, prompt record, per-annotation JSON record, elapsed time, and an evaluation CSV manifest under the local, Git-ignored `data/evaluation/` directory. The timer begins with the first prompt and ends on acceptance; model inference latency is measured separately. Ground truth is reserved for scoring and is not shown to guide prompt selection.

No model training, fine-tuning, or task-specific loss function is used. The model parameters are those of the downloaded SAM 2.1 checkpoint; prompt coordinates are supplied directly by the annotator. The high-level pipeline is:

COCO image and hidden reference mask ⟶ choose target object ⟶ human point/box prompts ⟶ local SAM 2.1 predictor ⟶ review and accept mask ⟶ persist mask, prompts, and elapsed time ⟶ compare with reference using IoU and Dice

Figure 1: Implemented annotation and evaluation pipeline. The ground-truth mask is used at the final scoring stage, not as prompt guidance.

4 Experimental Setup

4.1 Dataset

The data workflow selects 300 images from COCO 2017 val2017. Eligible images have between one and eight non-crowd instance annotations; candidates are sorted before selection with pseudorandom seed 42. The resulting subset contains 979 non-crowd annotations in 80 categories. Per-instance binary masks are exported for scoring. The selected image IDs are recorded in `docs/selected_image_ids.json`; download and regeneration instructions are in `data/README.md`. Images, annotation data, masks, and weights are local-only and ignored by Git. COCO source-image licensing must be checked before redistribution (COCO Consortium).

4.2 Baseline(s)

The planned primary baseline is manual polygon delineation of the same target instances, using a non-SAM annotation interface and the same timer definition. The baseline experiment was not run: no manual polygon masks or timing records were available when this report was prepared. Consequently, this report cannot confirm a baseline reproduction or compare against published baseline numbers. The existing COCO masks are reference annotations for scoring, not a measured manual-tool baseline.

4.3 Metrics & Protocol

For binary prediction mask P and reference mask G, intersection over union and Dice are computed as

$$\mathrm{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}, \qquad \mathrm{Dice}(P, G) = \frac{2|P \cap G|}{|P| + |G|}.$$

The implementation rejects mismatched mask dimensions and associates each prediction with its COCO annotation ID. Annotation time is the elapsed wall-clock time from the first prompt until acceptance; inference latency is reported separately. A valid paired comparison would require the same preselected target instances in both workflows, with ground truth withheld while each mask is created. No paired sample, repeated seed, or statistical test was completed. Therefore, no aggregate method metrics or time-saving estimate can be reported.

5 Results

The primary experiment remains incomplete. The evaluation directory contained no accepted predictions or timing manifest when checked for this report, and no manual-baseline records were available. Accordingly, the two method rows below explicitly report missing measurements rather than treating missing data as a zero score.

Table 1: Planned paired comparison. Neither workflow has scored observations; the SAM 2 smoke test below was not accepted or scored as an annotation.

Method	**Scored** N	**Mean IoU**	**Mean Dice**	**Annotation time**	**Runtime ev**
Manual polygon baseline	0	Not measured	Not measured	Not measured	Not measu
SAM 2.1 prompted tool	0	Not measured	Not measured	Not measured	One smoke-test re

The one smoke test used COCO image `000000000776.jpg` (428 by 640 pixels) and a positive point at $(130, 130)$ selected by visual inspection. With the SAM 2.1 Hiera-Tiny checkpoint on Apple MPS, the first API request returned a mask with confidence 0.954 in 1.56 seconds. This request included predictor initialization; it is not steady-state inference latency or human annotation time. Confidence is not IoU or Dice. No qualitative success/failure comparison is reported because the mask was not part of a reviewed, scored annotation trial.

No paired qualitative examples are reported. The available single smoke-test mask was not reviewed and scored against its target before this report.

Figure 2: Qualitative comparison unavailable: no verified paired manual and SAM 2 annotations were collected.

5.1 Ablation Studies

The planned prompt-type ablation would compare point prompting with box prompting on the same target instances, holding the checkpoint and acceptance protocol fixed. It was not performed, so no effect of prompt type can be estimated.

Table 2: Prompt-type ablation status. Missing values indicate that the experiment was not run.

Prompt condition	**Scored** N	**Mean IoU**	**Mean annotation time**
Positive point	0	Not measured	Not measured
Box	0	Not measured	Not measured

6 Discussion & Limitations

The completed contribution is a reproducible software and data path: the application can select COCO targets, accept interactive prompts, request local SAM 2 inference, and persist accepted masks with timing and mask metrics. The backend suite passed 17 tests, the frontend TypeScript production build and lint completed successfully, and a real one-image inference request verified that the downloaded checkpoint can run through the API on Apple MPS. These checks establish implementation behavior, not segmentation quality.

The central limitation is the absence of the planned paired human study. The manual polygon baseline was not collected, and no accepted SAM 2 masks were recorded in the evaluation manifest. The single smoke test has no ground-truth score and cannot support an accuracy or time-saving claim. In addition, the sample was designed as a small reproducibility subset rather

than a representative benchmark; no annotator study, inter-annotator analysis, uncertainty estimate, or per-category analysis was performed. The observed first-request latency includes model initialization and comes from one machine/device, so it should not be generalized to other hardware or steady-state requests.

The system also depends on local model weights and COCO assets that are excluded from version control. Reproduction requires downloading them with the documented scripts, installing the optional SAM 2 dependency, and matching the checkpoint to its configuration. The frontend and backend automate metric calculation but do not by themselves ensure that annotation prompts were chosen without seeing ground truth; this remains a protocol requirement for future trials. The results are therefore an honest implementation milestone, not evidence that SAM 2 is faster or more accurate than manual polygon labeling.

7 Conclusion

We built a local React/FastAPI annotation tool around frozen SAM 2.1 and a reproducible 300-image COCO subset, including an API path that stores masks and computes IoU, Dice, and annotation time. One real inference smoke test and automated software checks succeeded, but the paired manual baseline and scored prompted evaluation were not completed. No conclusion about quality or time savings is therefore justified; the next step is to collect blinded, same-instance annotations with the shared timer protocol.

Individual Contributions

Student A was assigned COCO data preparation and the manual polygon baseline; the baseline measurements remain incomplete. Student B implemented the local React/FastAPI annotation workflow, SAM 2 integration, metric persistence, and evaluation tooling.

References

Carsten Rother, Vladimir Kolmogorov, and Andrew Blake. Grabcut: Interactive foreground extraction using iterated graph cuts. In *ACM SIGGRAPH 2004 Papers*, pages 309–314, 2004. doi: 10.1145/1015706.1015720.

Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4015–4026, 2023. doi: 10.1109/ICCV51070.2023.00371.

Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Teng Ma, Hany Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Peter Vajda, Zongwei Lü, et al. SAM 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. URL https://arxiv.org/abs/2408.00714.

Xu Zhao, Wenchao Ding, Yongqi An, Yinglong Du, Tao Yu, Min Li, Ming Tang, and Jinqiao Wang. Fast segment anything. *arXiv preprint arXiv:2306.12156*, 2023. URL https://arxiv.org/abs/2306.12156.

Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *Computer Vision – ECCV 2014*, pages 740–755, 2014. doi: 10.1007/978-3-319-10602-1_48.

COCO Consortium. COCO terms of use. https://cocodataset.org/#termsofuse. Accessed October 31, 2026.

A Additional Results

The reproducible software checks at report time were 17 passing backend tests, successful Ruff checks, a successful frontend TypeScript/Vite production build, and a successful frontend lint run. The real inference smoke test returned a 428 by 640 mask with confidence 0.954; the first API request took 1.56 seconds on Apple MPS including predictor initialization. These are validation observations only and are not included as quality metrics.